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MIT SCHOOL OF COMPUTING

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BRAIN TUMOR SEGMENTATION & PROFILER

Deep learning platform for automated MRI brain tumor detection and slice-wise visualization

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What Users Say

- “How long until I get a result?”
- “Can I trust this tumor outline?”
- “Which scans do you need?”

What Users Think

- Manual tracing takes too long
- Are the borders precise enough?
- This should assist, not replace, judgment

Scan User

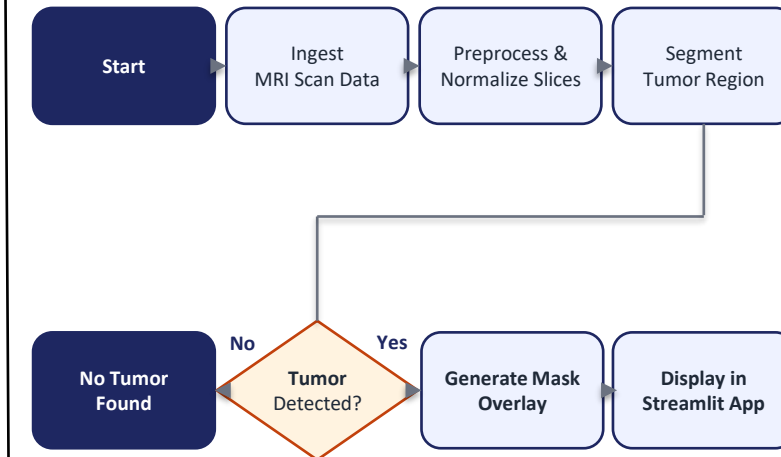
What Users Do

- Trace tumor boundaries slice by slice
- Cross-check outlines with colleagues
- Redo segmentation for every new scan

What Users Feel

- Fatigued by repetitive manual work
- Uncertain about consistency
- Reassured when output matches judgment

Sr. No	Requirement	Proposed Solution
1	Accurate Tumor Detection • Locate tumor region • Separate from healthy tissue • Work across MRI sequences	Fine-tuned MONAI segmentation model automatically detects and outlines the tumor region on each MRI slice.
2	Fast, Consistent Output • Manual tracing is slow • Results vary radiologist to radiologist	Automated inference produces a consistent mask in seconds, benchmarked against a measured Dice score.
3	Accessible Interface • Research models rarely usable directly	Streamlit app lets users upload a scan and instantly view slice-by-slice tumor mask overlays.



Problem statement

Manually outlining brain tumors in multi-parametric MRI scans is slow and varies between radiologists, delaying diagnosis and treatment planning for glioma patients. State-of-the-art deep learning segmentation models are typically trained as full 3D networks on the complete BraTS cohort, demanding heavy GPU infrastructure that makes them impractical for a lightweight, accessible academic prototype.

Proposed Solution

This project fine-tunes an existing pretrained MONAI segmentation model on a curated subset of BraTS MRI slices (T1ce and FLAIR) instead of training a 3D network from scratch. The model is wrapped in an interactive Streamlit app that lets a user upload a scan and instantly view the predicted tumor region overlaid on each slice, with a transparently reported Dice score.

Scope and Feasibility

The project covers automated tumor segmentation using a subset of the BraTS Challenge dataset (~100-150 patients, T1ce and FLAIR modalities) delivered through a lightweight Streamlit app. It targets binary tumor detection as its core deliverable, with multi-class sub-region segmentation as a future extension and runs entirely on free-tier cloud GPU resources.